

- Was using velocity, but plugged in speeds:
 - With V_x off lower limit 64,200, upper limit was 2000 (I think) & slope was - to I think

- 2000 (I think) - slope was - to 5 think
- The requirements for flight
- Pilot gain must be tuned properly each time
- I think it was 1000

• Let thrust = drag with basic trim

~~Thrust~~ = drag ~~A~~ ~~1~~ List = weights

$$T = \frac{1}{2} C_{0p} A v^2 + \frac{1}{2} C_{1p} A v^2 = mg$$

$$V_b = \sqrt{\frac{2mg}{C_{DPA}}} \quad V_L = + \sqrt{\frac{2mg}{C_{LPA}}}$$

* plug in the manipulated formula or the derivative value after passing a certain altitude

[Handwritten signature]

- We need the PID to perfectly curve to the changing desirable value because for drag, when ~~are~~ its overpowering, the aircraft will reverse.

- Is the force & moment outputting values that can be changed by this beam.

- Pressure points on

• Air flow thru is perfect gas $p = \rho RT$

$\rightarrow R$ is constant & for normal air it's $287 \frac{J}{(kg)(K)}$

$\frac{R}{C} = \frac{\text{excess power}}{W}$, can figure out the max velocity using the difference of IAP of engine that is ~~max~~ full throttle & what overcomes the aerodynamic drag.

"Skin friction drag: Flow separation is a friction that causes pressure drag due to separation from air flow

Airfoil A $\frac{1}{2}$ B $\frac{1}{4}$

The airflow velocity ~~on~~ is higher than B due to the design. This causes pressure ~~on the~~ ^{on the} top surface to have less pressure while on the B side there is more pressure pushing upward, then there's lift.

Airflow normally is a uniform flow but with the wing it causes incompressible flow thus the reason for lift

1 Drug Policy

* C_{Di} is parasite drag coefficient, this includes profile drag of wing, friction & pressure drag of the tail surface, fuselage, engine nacelles, landing gear, & other components exposed to airflow.

$$C_D = C_{D,0} + \frac{C_L^2}{\pi e A R} \quad \text{where } C_{D,0} = C_{D,\min} + \frac{(C_{L1} - C_{L2,\text{avg}})^2}{\pi e A R} \quad \text{where}$$

aircraft aligns with the relative wind.

Equation of Motion:

$$\Sigma F_{11} = T \cos \alpha_T - D - W \sin \theta = m \frac{dV}{dt}$$

$$\Sigma_i = L + T \sin \alpha - W \cos \theta = m \frac{v^2}{r_i}$$

- For Stabilization:

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Notes Rap
Aircraft